1 Geometric sequence

Remark 1.1 (Bernoulli's inequality). In HW4, Problem 3, you proved *Bernoulli's inequality*: for every $n \in \mathbb{N}$ and every real number $a \geq -1$,

$$(1+a)^n \geq 1 + na.$$

We will use this inequality to study geometric sequences.

Theorem 1.1. Let $\alpha \in \mathbb{C}$, and consider the geometric sequence $x_n = \alpha^n$. There are four cases

(i) If $\alpha = 1$, then $x_n = \alpha^n \to 1$ as $n \to \infty$.

(ii) If $|\alpha| < 1$, then $x_n = \alpha^n \to 0$ as $n \to \infty$.

(iii) If $|\alpha| = 1$ and $\alpha \neq 1$, then $x_n = \alpha^n$ diverges, while $|x_n| = 1$ for every $n \in \mathbb{N}$.

(iv) If $|\alpha| > 1$, then $|x_n| = |\alpha|^n \to +\infty$ as $n \to \infty$.

Proof. (i) Let $\alpha = 1$, then $x_n = \alpha^n = 1 \ \forall n \in \mathbb{N}$, and therefore x_n is a constant sequence that converges to 1.

(ii) Let $|\alpha| < 1$. If $\alpha = 0$, the sequence is constantly zero. Otherwise, $0 < |\alpha| < 1$, so $\frac{1}{|\alpha|} > 1$ and we can write for some $\theta > 0$:

$$\frac{1}{|\alpha|} = 1 + \theta, \qquad \theta > 0.$$

By Bernoulli's inequality,

$$\frac{1}{|\alpha|^n} = (1+\theta)^n \geq 1 + n\theta,$$

and therefore

$$0 \leq |\alpha^n| = |\alpha|^n \leq \frac{1}{1+n\theta}.$$

Let $\varepsilon > 0$. By the unboundedness of $\mathbb{N}$, choose $n_0(\varepsilon) \in \mathbb{N}$ such that

$$n_0(\varepsilon) > \frac{1/\varepsilon - 1}{\theta}.$$

Alternatively, we can choose $n_0(\varepsilon) = \max\left\{1, \left[\frac{1/\varepsilon - 1}{\theta}\right] + 1\right\}$. The maximum with 1 is needed because when $\varepsilon \geq 1$ the number $\frac{1/\varepsilon - 1}{\theta}$ is ≤ 0, and then $\left[\frac{1/\varepsilon - 1}{\theta}\right] + 1$ need not be a natural number.

Then we have that

$$\frac{1}{1 + n_0(\varepsilon)\theta} < \varepsilon.$$

For all $n \geq n_0(\varepsilon)$,

$$|\alpha^n - 0| \leq \frac{1}{1+n\theta} \leq \frac{1}{1+n_0(\varepsilon)\theta} < \varepsilon.$$

Hence $\alpha^n \to 0$. Recall from Proposition 3.2 from Lecture 15 that $\alpha^n \to 0$ if $|\alpha^n| \to 0$.

(iii) Let $|\alpha| = 1$ and $\alpha \neq 1$. We will prove that $x_n = \alpha^n$ diverges by contradiction. Suppose that $x_n = \alpha^n$ converges, i.e. $\exists a \in \mathbb{C}$ such that $\lim_{n\to\infty} \alpha^n = a$, which means

$$\forall \varepsilon > 0 \;\; \exists n_0(\varepsilon) \in \mathbb{N} \;\text{ such that }\; |\alpha^n - a| < \varepsilon \;\; \forall n \geq n_0(\varepsilon). \tag{*}$$

Let $\varepsilon = \frac{|\alpha-1|}{2}$ which is strictly positive since $\alpha \neq 1$. Then by (*) there exists $n_0(\varepsilon) \in \mathbb{N}$ such that

$$|\alpha^n - a| < \frac{|\alpha-1|}{2} \;\; \forall n \geq n_0(\varepsilon).$$

We also have $|\alpha^{n+1} - a| < \frac{|\alpha-1|}{2}$ for such n, since $n+1 > n$. Using the triangle inequality we then obtain

$$|\alpha^{n+1} - \alpha^n| = |\alpha^{n+1} - a + a - \alpha^n| \leq |\alpha^{n+1} - a| + |a - \alpha^n| < |\alpha - 1|.$$

On the other hand, since $|\alpha| = 1$ we have

$$|\alpha^{n+1} - \alpha^n| = |\alpha^n(\alpha - 1)| = |\alpha|^n |\alpha - 1| = |\alpha - 1|.$$

We thus showed

$$|\alpha - 1| = |\alpha^{n+1} - \alpha^n| < |\alpha - 1|,$$

which is a contradiction. Thus $x_n = \alpha^n$ does not converge to any complex number. All its terms lie on the unit circle.

(iv) Let $|\alpha| > 1$. Write $|\alpha| = 1 + \theta$ with $\theta > 0$. By Bernoulli's inequality,

$$|\alpha^n| = |\alpha|^n = (1+\theta)^n \geq 1 + n\theta.$$

Given any $M > 0$, by the unboundedness of $\mathbb{N}$ we can choose $n_0(M) \in \mathbb{N}$ such that $n_0(M) > (M-1)/\theta$.

Alternatively, we can choose $n_0(M) = \max\left\{1, \left[\frac{M-1}{\theta}\right] + 1\right\}$. As in case (ii), the maximum with 1 is needed because when $M \leq 1$ the number $\frac{M-1}{\theta}$ is ≤ 0.

Then we will have that

$$1 + n_0(M)\theta > M.$$

For every $n \geq n_0(M)$,

$$|\alpha^n| \geq 1 + n\theta \geq 1 + n_0(M)\theta > M.$$

Thus $|\alpha^n| \to +\infty$.

□

Remark 1.2 (How to choose n_0). In cases (ii) and (iv), the choice of the threshold index n_0 comes from working backwards on scratch paper, using the estimates obtained from Bernoulli's inequality.
In case (ii), where $0 < |\alpha| < 1$, we have

$$|\alpha^n| \leq \frac{1}{1+n\theta}, \qquad \text{where } \theta = \frac{1}{|\alpha|} - 1 > 0.$$

To ensure that $|\alpha^n| < \varepsilon$, it suffices to have

$$\frac{1}{1+n\theta} < \varepsilon \iff 1 + n\theta > \frac{1}{\varepsilon} \iff n > \frac{1/\varepsilon - 1}{\theta}.$$

This suggests choosing $n_0(\varepsilon) \in \mathbb{N}$ larger than $\frac{1/\varepsilon - 1}{\theta}$.
In case (iv), where $|\alpha| > 1$, we write $|\alpha| = 1 + \theta$ with $\theta > 0$. Bernoulli's inequality gives

$$|\alpha^n| \geq 1 + n\theta.$$

To show that $|\alpha^n| \to +\infty$, we want $|\alpha^n| > M$ for every sufficiently large n, where $M > 0$ is arbitrary. It is therefore sufficient to arrange that

$$1 + n\theta > M \iff n\theta > M - 1 \iff n > \frac{M-1}{\theta}.$$

This suggests choosing any $n_0(M) \in \mathbb{N}$ satisfying $n_0(M) > (M-1)/\theta$. Then for every $n \geq n_0(M)$, $|\alpha^n| \geq 1 + n\theta > M$.
In both cases, we discover the appropriate choice of n_0 by working backwards, but present the proof forwards.

Remark 1.3 (Case (iv) from case (ii)). Once case (ii) is proved, case (iv) follows almost for free. Suppose $|\alpha| > 1$ and set $\beta = 1/\alpha$. Then $0 < |\beta| < 1$, so by case (ii),

$$|\beta|^n = \frac{1}{|\alpha|^n} \longrightarrow 0.$$

Now use the following general fact: *if $y_n > 0$ for every n and $y_n \to 0$, then $1/y_n \to +\infty$.* Indeed, given $M > 0$, apply the definition of convergence with $\varepsilon = 1/M$: there is $n_0 \in \mathbb{N}$ such that $0 < y_n < 1/M$ for all $n \geq n_0$, that is, $1/y_n > M$ for all $n \geq n_0$. Applying this with $y_n = |\beta|^n$ gives $|\alpha|^n = 1/|\beta|^n \to +\infty$, which is case (iv). We gave a direct proof of case (iv) above because it is good practice in choosing n_0.

Remark 1.4 (A different approach using logarithms). Once we have studied the exponential function and its inverse, the logarithm, we will have another way to prove case (ii). Suppose $0 < |\alpha| < 1$ and set $a = |\alpha|$. To show that $\alpha^n \to 0$, we want

$$a^n < \varepsilon.$$

Taking logarithms on both sides of the inequality (one can do this because the logarithm is what we call an increasing function), we obtain

$$n \ln a < \ln \varepsilon \iff n > \frac{\ln \varepsilon}{\ln a},$$

where the inequality reverses because $\ln a < 0$. Thus we could choose any $n_0(\varepsilon) \in \mathbb{N}$ satisfying

$$n_0(\varepsilon) > \frac{\ln \varepsilon}{\ln a}.$$

We will introduce the number e in the next lecture.

Examples

(i) $x_n = (-1)^n$ diverges. Indeed, we have $\alpha = -1$, so $\alpha \neq 1$ and $|\alpha| = 1$, which is case (iii). Thus $x_n = (-1)^n$ diverges.

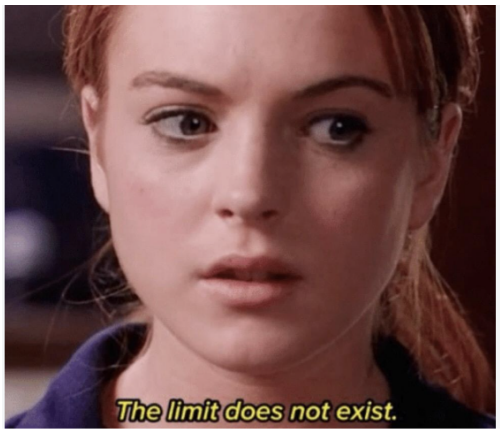


Figure 1: The limit of $(-1)^n$ does not exist, and the above proof is the second one of this fact that we have seen.

(ii) $x_n = (1/2 + i/4)^n$ converges to 0. In this case we have

$$\alpha = \frac{1}{2} + \frac{i}{4} \;\Rightarrow\; |\alpha| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^2} = \sqrt{\frac{5}{16}} < 1,$$

which is case (ii) of Theorem 1.1.

Exercise

Using Theorem 1.1, determine whether each of the following sequences converges.

(i) $x_n = \left(\frac{1+i}{2}\right)^n$.

(ii) $x_n = i^n$.

(iii) $x_n = 2^n$.

2 Some other basic limits

Proposition 2.1. If $a > 0$, then the sequence $x_n = \sqrt[n]{a} \to 1$.

Proof. Case 1: $a > 1$. Then $\sqrt[n]{a} > 1$ for every $n \in \mathbb{N}$. Define

$$\theta_n = \sqrt[n]{a} - 1 = x_n - 1.$$

Observe that $\theta_n > 0$ for every $n \in \mathbb{N}$. If we show that $\theta_n \to 0$, we obtain the desired result: $x_n = 1 + \theta_n \to 1$. Since $\sqrt[n]{a} = 1 + \theta_n$, we can write using Bernoulli's inequality

$$a = (1 + \theta_n)^n \geq 1 + n\theta_n > n\theta_n.$$

It follows that

$$0 < \theta_n < \frac{a}{n},$$

and by the squeeze theorem we conclude that $\theta_n \to 0$. Consequently, $x_n = 1 + \theta_n \to 1$.

Case 2: $0 < a < 1$. Then $1/a > 1$. By Case 1,

$$\frac{1}{x_n} = \frac{1}{\sqrt[n]{a}} = \sqrt[n]{\frac{1}{a}} \to 1 \neq 0.$$

Consequently, $x_n \to 1$.

Case 3: $a = 1$. Then $x_n = \sqrt[n]{1} = 1$ for every $n \in \mathbb{N}$, so $x_n \to 1$. □

Proposition 2.2. The sequence $x_n = \sqrt[n]{n} \to 1$.

Proof. The trick is to apply Bernoulli's inequality to $\sqrt{n}$ instead of n. Since $n \geq 1$, we have $\sqrt[2n]{n} \geq 1$, so we can write

$$\sqrt[2n]{n} = 1 + \delta_n, \qquad \delta_n \geq 0.$$

Raising to the power n and using Bernoulli's inequality,

$$\sqrt{n} = (1 + \delta_n)^n \geq 1 + n\delta_n > n\delta_n,$$

and therefore

$$0 \leq \delta_n < \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}}.$$

Since $1/n \to 0$ and square roots preserve limits, $1/\sqrt{n} \to 0$, so $\delta_n \to 0$ by the squeeze theorem. Finally,

$$\sqrt[n]{n} = \left(\sqrt[2n]{n}\right)^2 = (1+\delta_n)^2 \longrightarrow 1.$$

□

3 Root and ratio criteria of convergence

Proposition 3.1. [**Ratio criterion**] Let (a_n) be a sequence of nonzero complex numbers.

(i) If $\left|\frac{a_{n+1}}{a_n}\right| \to \ell$ with $\ell < 1$, then $a_n \to 0$.

(ii) If $\left|\frac{a_{n+1}}{a_n}\right| \to \ell$ with $\ell > 1$, then $|a_n| \to +\infty$.

(iii) If $\left|\frac{a_{n+1}}{a_n}\right| \to +\infty$, then $|a_n| \to +\infty$.

Proof. (i) Set $\rho = \frac{1+\ell}{2}$, so that $\ell < \rho < 1$.

Since $\left|\frac{a_{n+1}}{a_n}\right| \to \ell$, there is $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$,

$$\left|\frac{a_{n+1}}{a_n}\right| < \rho.$$

We obtain this by applying the definition of convergence with $\varepsilon = \rho - \ell > 0$: for all $n \geq n_0$ we get $\left|\frac{a_{n+1}}{a_n}\right| < \ell + \varepsilon = \rho$. The idea is that if a sequence converges to ℓ and $\ell < \rho$, then all but finitely many terms of the sequence are smaller than ρ.

For every $n \geq n_0$, we have

$$|a_{n+1}| \leq \rho |a_n|.$$

Applying this inequality repeatedly, we obtain

$$\begin{aligned}
|a_{n_0+1}| &\leq \rho |a_{n_0}|, \\
|a_{n_0+2}| &\leq \rho |a_{n_0+1}| \leq \rho^2 |a_{n_0}|, \\
|a_{n_0+3}| &\leq \rho |a_{n_0+2}| \leq \rho^3 |a_{n_0}|, \\
&\vdots
\end{aligned}$$

Continuing in this way, for every $n > n_0$ we obtain

$$|a_n| \leq |a_{n_0}| \rho^{n-n_0}.$$

Since $0 < \rho < 1$, we know that $\rho^{n-n_0} \to 0$ by Theorem 1.1. Thus, by the squeeze theorem, $|a_n| \to 0$, and consequently $a_n \to 0$.

(ii) Set $\mu = \frac{1+\ell}{2}$, so that $1 < \mu < \ell$. Since $\left|\frac{a_{n+1}}{a_n}\right| \to \ell$, there is $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$,

$$\left|\frac{a_{n+1}}{a_n}\right| > \mu.$$

As in part (i), we obtain this by applying the definition of convergence with $\varepsilon = \ell - \mu > 0$: for all $n \geq n_0$ we get $\left|\frac{a_{n+1}}{a_n}\right| > \ell - \varepsilon = \mu$. If a sequence converges to ℓ and $\ell > \mu$, then all but finitely many terms of the sequence are larger than μ.

For every $n \geq n_0$, we have

$$|a_{n+1}| \geq \mu |a_n|.$$

Applying this inequality repeatedly, we obtain

$$\begin{aligned}
&|a_{n_0+1}| \geq \mu |a_{n_0}|, \\
&|a_{n_0+2}| \geq \mu |a_{n_0+1}| \geq \mu^2 |a_{n_0}|, \\
&|a_{n_0+3}| \geq \mu |a_{n_0+2}| \geq \mu^3 |a_{n_0}|, \\
&\qquad \vdots
\end{aligned}$$

Continuing in this way, for every $n > n_0$ we obtain

$$|a_n| \geq |a_{n_0}| \mu^{n-n_0}.$$

Since $\mu > 1$, we know that $\mu^{n-n_0} \to +\infty$. Moreover, $|a_{n_0}| > 0$ since the terms of the sequence are nonzero. Therefore,

$$|a_{n_0}| \mu^{n-n_0} \longrightarrow +\infty.$$

Consequently, $|a_n| \to +\infty$.

(iii) Since $\left|\frac{a_{n+1}}{a_n}\right| \to +\infty$, applying the definition of divergence to $+\infty$ with $M = 2$ gives $n_0 \in \mathbb{N}$ such that

$$\left|\frac{a_{n+1}}{a_n}\right| > 2 \qquad \text{for all } n \geq n_0.$$

The argument of part (ii) with $\mu = 2$ now gives $|a_n| \geq |a_{n_0}|\, 2^{n-n_0}$ for every $n > n_0$, and hence $|a_n| \to +\infty$.

□

Remark 3.1. When $\ell = 1$, the ratio criterion gives no conclusion: the sequences $a_n = n$, $a_n = \frac{1}{n}$, and $a_n = 1$ all satisfy that $|\frac{a_{n+1}}{a_n}| \to 1$, but have different limiting behavior.

Remark 3.2. The idea behind the ratio criterion is to compare a sequence with a geometric sequence. Recall that for a geometric sequence $x_n = \alpha^n$, the ratio of consecutive terms is constant:

$$\frac{x_{n+1}}{x_n} = \alpha.$$

By Theorem 1.1, if $|\alpha| < 1$, then $x_n \to 0$, while if $|\alpha| > 1$, then $|x_n| \to +\infty$.
For a general sequence (a_n), the ratio of consecutive terms need not be constant. Nevertheless, if these ratios are eventually bounded above by some $\rho < 1$, or bounded below in modulus by some $\mu > 1$, we can compare $(|a_n|)$ with a geometric sequence to determine its limiting behavior.

Proposition 3.2. **[Root criterion]** Let (a_n) be a sequence of complex numbers, and suppose that

$$\sqrt[n]{|a_n|} \longrightarrow \ell.$$

(i) If $\ell < 1$, then $a_n \to 0$.

(ii) If $\ell > 1$, then $|a_n| \to +\infty$.

Proof. (i) Set $\rho = (1 + \ell)/2$, so that $\ell < \rho < 1$. Applying the definition of convergence with $\varepsilon = \rho - \ell$, for all sufficiently large n,

$$\sqrt[n]{|a_n|} < \rho, \qquad \text{hence} \qquad 0 \le |a_n| < \rho^n.$$

Because $\rho^n \to 0$, we conclude that $a_n \to 0$.

(ii) Set $\mu = (1+\ell)/2$, so that $1 < \mu < \ell$. Applying the definition of convergence with $\varepsilon = \ell - \mu$, for all sufficiently large n,

$$\sqrt[n]{|a_n|} > \mu, \qquad \text{hence} \qquad |a_n| > \mu^n \longrightarrow +\infty.$$

□

Remark 3.3. When $\ell = 1$, the root criterion gives no conclusion. Indeed, $\sqrt[n]{n} \to 1$, $\sqrt[n]{1/n} \to 1$, and $\sqrt[n]{1} \to 1$, although the corresponding sequences have different limiting behavior.

4 Computing limits of sequences

Predicting limits based on growth rates

Before trying to prove that a sequence converges (or diverges), it can be useful to *predict* its behavior by comparing the growth of the expressions involved.

Exercise 1. For each sequence, first predict its behavior by comparing the growth rates of the numerator and denominator. Then justify your conjecture rigorously.

(i) $a_n = \dfrac{3n^3 + 2n}{n^2 + 1}$.

(ii) $b_n = \dfrac{2n^3 - n + 1}{5n^3 + 2n^2}$.

(iii) $u_n = \dfrac{n^2}{2^n}$.

(iv) $v_n = \dfrac{n!}{3^n}$.

What do these four examples suggest about the relative growth of polynomials, exponentials, and factorials? Which arguments turn that intuition into a proof?

Additional practice (optional)

For each sequence below, determine whether it converges. If it converges, find its limit; if it diverges, explain why. These problems are for independent practice and are not part of the in-class group activity.

(i) $g_n = \sqrt{n+\sqrt{n}} - \sqrt{n}$. *Hint: Multiply and divide by the conjugate expression* $\sqrt{n+\sqrt{n}} + \sqrt{n}$.

(ii) $h_n = \left(\sqrt[n]{n} - 1\right)^n$.

(iii) $j_n = \dfrac{5^n + n}{6^n - n}$.